Path-Space Variational Bridges: A Graphical-Model View of Diffusions, Rectified Flows, and Poisson Flows

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Abstract

Diffusion models, score-based stochastic differential equations, flow matching, rectified flows, Schrödinger bridges, and Poisson-flow generative models are often presented as separate generative paradigms. This preliminary note proposes a common mathematical object behind them: a *path-space variational bridge*. The construction has four ingredients: an endpoint coupling between a base distribution and the data distribution, a conditional bridge measure on latent paths, a Markovian projection that converts non-causal bridges into causal generative dynamics, and a probability-current gauge that decides whether the same marginal path is represented as an ordinary differential equation or a stochastic differential equation. In this view, diffusion models are hierarchical/path-space variational autoencoders with a noisy encoder bridge; rectified flow and flow matching are zero-temperature or deterministic-gauge projections of a chosen bridge; Schrödinger bridges are entropy-regularized Brownian bridge instances; and Poisson/electrostatic flows are harmonic-field bridge geometries in augmented space. We formalize this viewpoint, derive the corresponding matching losses, state a gauge equivalence between ODE and SDE realizations, and outline research directions suggested by recent work on stochastic interpolants, latent stochastic interpolants, bridge matching, and field matching.

Status. This is a conceptual and mathematical preliminary draft. It makes no empirical claim. The purpose is to isolate a general latent graphical model and to make explicit which parts of modern generative modeling are choices of bridge geometry, which parts are variational inference, and which parts are gauge choices for the same probability current.

1 Introduction

Many recent generative models learn a continuous transformation from a simple distribution to a data distribution. Diffusion probabilistic models learn to reverse a noising Markov chain and are trained through a variational bound (1; 2; 4). Score-based SDE models describe the same idea in continuous time and derive both reverse-time SDE samplers and equivalent probability-flow ODEs (3). Flow matching and rectified flow instead learn an ODE vector field by regressing conditional velocities of prescribed probability paths (7; 8; 9). Stochastic interpolants provide a broad bridge construction that recovers deterministic flows and stochastic diffusions from the same interpolating process (11; 10). Schrödinger bridge methods replace deterministic transport by entropy-regularized stochastic transport (12; 13). Poisson-flow generative models and recent electrostatic field-matching methods take a different route: data distributions become charges or capacitor plates in an augmented space, and generation follows field lines (16; 17; 18; 19; 20).

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The working hypothesis of this note is that the common structure is not a particular SDE, ODE, score, velocity, noise schedule, or electric field. The common structure is a *latent path graphical model*. A modeler first specifies how a simple endpoint and a data endpoint are coupled. Then the modeler specifies a conditional path law, or bridge, between those endpoints. This produces a teacher path measure. The learned generator is a causal Markov process obtained by projecting that teacher path measure onto a Markov family. Finally, the same marginal evolution can be represented by different stochastic gauges: a deterministic probability-flow ODE, a noisy SDE with a score correction, or an augmented-space field flow.

This viewpoint suggests the following slogan:

> *Diffusion and rectified flow are not fundamentally different models; they are different gauges and bridge choices inside the same path-space variational graphical model.*

1.1 Contributions of this preliminary note

1. We define a *path-space variational bridge* using an endpoint coupling, a conditional bridge kernel, and a learned Markov decoder process.

2. We show how flow-matching and rectified-flow losses arise as Markovian projections of non-causal bridge velocities.

3. We state a probability-current gauge theorem: if a marginal density path solves a continuity equation, then an infinite family of SDEs with different diffusion matrices realizes the same marginals by changing the drift with a score/divergence correction.

4. We locate DDPM/score SDEs, flow matching, rectified flow, Brownian/Schrödinger bridges, stochastic interpolants, and Poisson/electrostatic flows as specializations.

5. We outline open problems toward a fully learnable VAE-like theory of bridge geometry, coupling, and stochastic gauge.

2 Related work and active directions

Diffusions as variational latent-variable models. Diffusion probabilistic models were introduced as multi-step latent-variable generative models trained through variational inference (1; 2). Variational diffusion models further sharpened the continuous-time variational bound and the role of signal-to-noise schedules (4). A continuous-time variational perspective also connects diffusion models, score matching, hierarchical VAEs, and normalizing flows (5). Optimal-control formulations interpret diffusion training as KL minimization between path measures and derive ELBOs from stochastic control arguments (6).

Flows, rectification, and stochastic interpolants. Flow matching trains continuous normalizing flows without simulating ODEs during training by regressing vector fields of fixed conditional probability paths (7). Rectified flow learns ODEs that follow straight-line couplings and can use reflow to straighten trajectories (8). Conditional flow matching and optimal-transport CFM generalize the construction to arbitrary sources and minibatch couplings (9). Stochastic interpolants and related normalizing-flow constructions explicitly

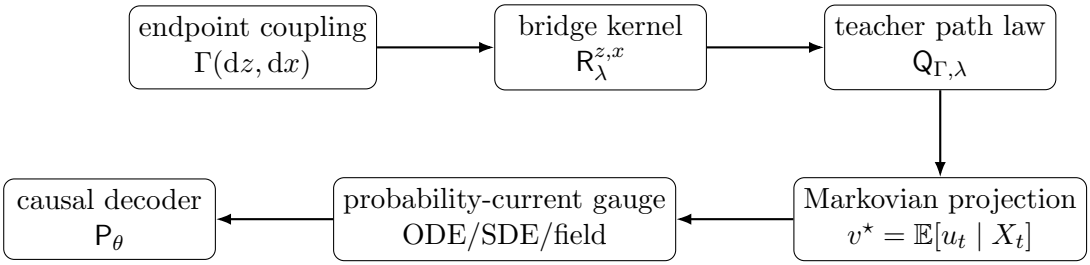

Figure 1: Path-space variational bridge: choose endpoint coupling and bridge geometry, aggregate a teacher path measure, project it to a causal Markov process, then choose a stochastic gauge for the same probability current.

unify flows and diffusions through interpolating stochastic processes and probability currents (10; 11).

Bridge problems. Schrödinger bridges solve an entropy-regularized dynamic transport problem over path measures. Recent methods scale bridge computation by combining score and flow matching, iterative Markovian fitting, and matching objectives (12; 13; 14). Diffusion flow matching gives KL guarantees for Brownian bridge choices and explicitly describes the procedure as choosing a coupling plus a deterministic or stochastic bridge, then learning the drift of a Markovian projection (15).

Poisson, electrostatic, and field matching. Poisson Flow Generative Models interpret data as electric charges on a hyperplane of an augmented space and sample by following the induced electric field (16). PFGM++ introduces an $N+D$-dimensional family that recovers PFGM at $D = 1$ and diffusion models as $D \to \infty$ (17). Electrostatic Field Matching places source and target distributions on capacitor plates and learns the field between them (18). Interaction Field Matching generalizes beyond electrostatics (19). Recent work on flow/field matching duality studies when conditional flow matching and interaction-field matching describe the same dynamics, and when field models are strictly more expressive (20).

The closest active direction. Latent Stochastic Interpolants are especially close to the present viewpoint: they jointly learn an encoder, decoder, and latent stochastic interpolant through a continuous-time ELBO (21). The difference in emphasis is that the present note treats the bridge itself as a graphical-model primitive and adds the probability-current gauge as a separate degree of freedom.

3 Path-space variational bridge

Let $\mathcal{M}$ be the state space, usually $\mathbb{R}^d$ or an augmented space such as $\mathbb{R}^d \times \mathbb{R}^D$. Let $\mathcal{X} = C([0,1], \mathcal{M})$ be the path space. We denote by $X_t : \mathcal{X} \to \mathcal{M}$ the coordinate process. Let π_0 be a simple base distribution and let $\pi_1 = \pi_{\text{data}}$ be the data distribution.

Definition 1 (Endpoint coupling)**.** *An endpoint coupling is a probability measure*

$$\Gamma(\mathrm{d}z, \mathrm{d}x) \in \Pi(\pi_0, \pi_1), \tag{3.1}$$

where $\Pi(\pi_0, \pi_1)$ is the set of couplings whose first marginal is π_0 and second marginal is π_1.

Examples include independent coupling, minibatch optimal-transport coupling, entropic optimal-transport coupling, paired data coupling, or a learned encoder coupling of the form

$$\Gamma_\phi(\mathrm{d}z, \mathrm{d}x) = q_\phi(\mathrm{d}z \mid x)\pi_{\mathrm{data}}(\mathrm{d}x). \tag{3.2}$$

The last case is the most VAE-like: the base-side endpoint is not merely noise but an inferred latent variable.

Definition 2 (Conditional bridge kernel)**.** *For each endpoint pair (z, x), a bridge kernel is a path-space probability measure*

$$\mathsf{R}_\lambda^{z,x}(\mathrm{d}X_{0:1}), \qquad X_0 = z, \quad X_1 = x, \tag{3.3}$$

where λ denotes bridge hyperparameters or learnable bridge parameters.

The bridge kernel is where the modeler encodes geometry. It may be deterministic, Brownian, Schrödinger, Hamiltonian, harmonic, or learned. Given Γ and R_λ, define the aggregated teacher path law

$$\mathsf{Q}_{\Gamma,\lambda}(\mathrm{d}X_{0:1}) = \int \mathsf{R}_\lambda^{z,x}(\mathrm{d}X_{0:1})\, \Gamma(\mathrm{d}z, \mathrm{d}x). \tag{3.4}$$

Definition 3 (Path-space variational bridge)**.** *A path-space variational bridge consists of*

$$(\Gamma_\phi, \mathsf{R}_\lambda, \mathsf{P}_\theta, p_\psi), \tag{3.5}$$

where Γ_ϕ is an endpoint inference coupling, R_λ is a conditional bridge, P_θ is a causal Markov path law initialized from π_0, and $p_\psi(x \mid y_1)$ is an optional observation decoder.

The simplest generative model uses Y_1 itself as the sample. A more VAE-like model decodes from Y_1 through $p_\psi(x \mid Y_1)$. The continuous-time ELBO has the formal form

$$\log p_{\theta,\psi}(x) \geq \mathbb{E}_{\mathsf{Q}_\phi(\mathrm{d}Y_{0:1}|x)} \left[\log p_\psi(x \mid Y_1) - \log \frac{\mathrm{d}\mathsf{Q}_\phi(\cdot \mid x)}{\mathrm{d}\mathsf{P}_\theta}(Y_{0:1}) \right], \tag{3.6}$$

whenever the conditional inference path law is absolutely continuous with respect to the generative path law. If both laws share a diffusion covariance and differ only in drift, Girsanov's theorem turns the path KL term into an integrated drift-matching cost. This is the continuous-time analog of a hierarchical VAE with infinitely many latent layers.

4 Markovian projection and matching losses

Assume for the moment that $\mathsf{Q} = \mathsf{Q}_{\Gamma,\lambda}$ is supported on differentiable paths, or more generally that it has a local velocity or drift target u_t satisfying

$$\frac{\mathrm{d}}{\mathrm{d}t} f(X_t) = \nabla f(X_t) \cdot u_t \tag{4.1}$$

for smooth test functions in the deterministic case. We want a causal Markov ODE

$$\mathrm{d}Y_t = v_\theta(Y_t, t)\, \mathrm{d}t, \qquad Y_0 \sim \pi_0, \tag{4.2}$$

that has the same one-time marginals as the teacher, or at least approximates them.

Proposition 1 (Projection loss)**.** *For a fixed teacher path law* Q*, the population minimizer of*

$$\mathcal{L}_{\mathrm{FM}}(v_\theta) = \int_0^1 \mathbb{E}_{\mathsf{Q}}\left[\|u_t - v_\theta(X_t, t)\|^2\right] \mathrm{d}t \tag{4.3}$$

is

$$v^\star(x,t) = \mathbb{E}_{\mathsf{Q}}\left[u_t \mid X_t = x\right]. \tag{4.4}$$

Thus, flow matching is a Markovian projection of a non-causal bridge. The non-causal bridge may know both endpoints (Z, X); the learned generator can only look at the current state and time. The conditional expectation in Eq. (4.4) is exactly the causalization step.

For an Itô teacher process

$$\mathrm{d}X_t = \beta_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}B_t, \tag{4.5}$$

with possibly non-Markov adapted drift β_t, the Markovian projected drift is similarly

$$b^\star(x,t) = \mathbb{E}_{\mathsf{Q}}[\beta_t \mid X_t = x], \tag{4.6}$$

under standard mimicking assumptions. Diffusion flow matching with Brownian bridges can be read in precisely this way (15).

5 Probability-current gauge

The central object is the marginal density path ρ_t and its probability current J_t. If an ODE velocity v_t transports ρ_t, then

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \qquad J_t = \rho_t v_t. \tag{5.1}$$

For an SDE

$$\mathrm{d}X_t = b_t(X_t)\,\mathrm{d}t + \sigma_t(X_t)\,\mathrm{d}B_t, \qquad a_t = \sigma_t \sigma_t^\top, \tag{5.2}$$

the Fokker-Planck equation can be written as a continuity equation

$$\partial_t \rho_t + \nabla \cdot J_t = 0, \qquad J_{t,i} = \rho_t b_{t,i} - \frac{1}{2}\sum_j \partial_j (a_{t,ij}\rho_t). \tag{5.3}$$

Theorem 1 (Probability-current gauge)**.** *Suppose* ρ_t *and* v_t *solve Eq.* (5.1)*. Let* $a_t(x) \succeq 0$ *be a smooth diffusion matrix. Define*

$$b_i^{(a)}(x,t) = v_i(x,t) + \frac{1}{2\rho_t(x)} \sum_j \partial_j \left(a_{ij}(x,t)\rho_t(x)\right). \tag{5.4}$$

Then the SDE with drift $b^{(a)}$ *and diffusion matrix* a *has the same marginal density path* ρ_t *as the ODE velocity* v_t*, provided the Fokker-Planck equation is well posed. If* $a_t(x) = g(t)^2 I$*, then*

$$b^{(a)}(x,t) = v_t(x) + \frac{g(t)^2}{2} \nabla \log \rho_t(x). \tag{5.5}$$

This theorem separates *transport* from *gauge*. The marginal path is determined by $J_t = \rho_t v_t$. A deterministic flow uses $a = 0$. A diffusion uses $a \neq 0$ and compensates with a score term. Thus the score is not an independent ontology; it is the correction needed to express the same probability current in a noisy gauge.

Model family	Coupling/bridge choice	Learned object	Gauge/view
DDPM / score SDE	fixed Gaussian noising bridge from data to noise	score or denoiser	stochastic gauge, variational path ELBO
Flow matching	conditional probability path between endpoints	velocity field	ODE gauge
Rectified flow	straight-line bridge $X_t = (1-t)Z + tX$	$\mathbb{E}[X - Z \mid X_t]$	deterministic bridge and ODE gauge
Brownian bridge / DFM	Brownian bridge conditioned on endpoints	Markovian projected drift	stochastic bridge, SDE gauge
Schrödinger bridge	KL-optimal path measure relative to Brownian reference	forward/backward drifts or scores	entropy-regularized stochastic transport
PFGM / EFM / IFM	harmonic or interaction field in augmented space	field lines or interaction field	augmented-space field gauge

Table 1: Several generative paradigms as choices inside the same path-space bridge template.

6 Instantiations

6.1 Diffusion models as path-space VAEs

A discrete DDPM defines a fixed inference/noising process

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}) \tag{6.1}$$

and a learned reverse Markov chain

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t). \tag{6.2}$$

The latent variables are the entire chain $(x_1, \ldots, x_T)$. The training bound is a variational bound on $\log p_\theta(x_0)$ (2; 4). In the path-space bridge language, the inference bridge starts at data and ends at a tractable base distribution; the decoder reverses this path. The graphical model is therefore a hierarchical VAE with a fixed, very deep encoder.

In continuous time, the score-based SDE formulation chooses a forward noising SDE and uses the time-reversal formula to define a generative reverse SDE whose drift depends on $\nabla \log \rho_t$ (3). The probability-flow ODE is the zero-noise representation of the same marginal path.

6.2 Rectified flow and flow matching

For straight-line rectified flow, sample $(Z, X) \sim \Gamma$ and define

$$X_t = (1-t)Z + tX, \qquad u_t = X - Z. \tag{6.3}$$

The projection loss becomes

$$\mathcal{L}_{\mathrm{RF}}(v_\theta) = \mathbb{E}_{t,Z,X}\left[\|X - Z - v_\theta((1-t)Z + tX, t)\|^2\right]. \tag{6.4}$$

The learned ODE velocity is the causal projection

$$v^\star(x,t) = \mathbb{E}[X - Z \mid (1-t)Z + tX = x]. \tag{6.5}$$

Thus, rectified flow is not essentially about straight lines; it is about projecting a chosen non-causal bridge into a causal Markov flow. Straightness is a bridge prior.

General flow matching replaces the deterministic line with a family of conditional probability paths $p_t(\cdot \mid z, x)$ and conditional vector fields. It still minimizes a regression loss whose population minimizer is the marginal vector field (7; 9).

6.3 Brownian bridges and Schrödinger bridges

A Brownian bridge from z to x with variance parameter ϵ can be written informally as

$$X_t = (1-t)z + tx + \sqrt{\epsilon}\,(B_t - tB_1). \tag{6.6}$$

This bridge has endpoint noise geometry rather than straight-line deterministic geometry. Aggregating Eq. (6.6) over a coupling Γ yields a path law whose Markovian projection gives a diffusion model. Diffusion flow matching analyzes precisely such bridge-induced path measures and the KL error of their learned projections (15).

Schrödinger bridges go one step further: instead of fixing the Brownian bridge kernel, they search for the path measure closest in KL to a Brownian reference among all path measures with prescribed endpoints. This makes the bridge geometry adaptive but entropy-regularized. Recent matching algorithms seek scalable approximations to these bridges (12; 13; 14).

6.4 Poisson and electrostatic field flows

Poisson Flow Generative Models embed d-dimensional data into $\mathbb{R}^d \times \mathbb{R}$ by placing data charges on the $z = 0$ hyperplane. The potential Φ solves a Poisson equation, and the electric field

$$E = -\nabla\Phi \tag{6.7}$$

defines field-line transport from a simple distribution on an augmented surface to the data hyperplane (16). PFGM++ extends this to $\mathbb{R}^d \times \mathbb{R}^D$, where $D = 1$ recovers PFGM and $D \to \infty$ recovers diffusion-model behavior (17).

In the present language, Poisson and electrostatic methods choose a bridge geometry through a harmonic or interaction field rather than through Brownian motion or linear interpolation. The bridge is not merely a time-indexed interpolation in data space. It is an augmented-space geometry whose field lines induce endpoint couplings and paths.

6.5 Gaussian bridge: score, noise, denoiser, and velocity are reparameterizations

Consider the Gaussian conditional path

$$X_t = \alpha_t X + \sigma_t \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I), \tag{6.8}$$

where $X \sim \pi_{\text{data}}$. Let ρ_t be the density of X_t and $s_t(x) = \nabla \log \rho_t(x)$. By Tweedie-style conditioning,

$$\mathbb{E}[\varepsilon \mid X_t = x] = -\sigma_t s_t(x). \tag{6.9}$$

The velocity target is

$$v^\star(x,t) = \mathbb{E}[\dot{\alpha}_t X + \dot{\sigma}_t \varepsilon \mid X_t = x] \tag{6.10}$$

$$= \frac{\dot{\alpha}_t}{\alpha_t} x + \left(\frac{\dot{\alpha}_t \sigma_t^2}{\alpha_t} - \dot{\sigma}_t \sigma_t \right) s_t(x). \tag{6.11}$$

Hence denoising, noise prediction, score prediction, and velocity prediction are linear reparameterizations once the bridge schedule (α_t, σ_t) is fixed. This explains why diffusion and Gaussian flow matching are often interchangeable at the level of learned information, even if their samplers and parameterizations differ.

7 Training objectives suggested by the framework

The framework separates four learning choices.

1. Learn or choose the endpoint coupling. A simple choice is independent coupling $\pi_0 \otimes \pi_{\text{data}}$. A transport-biased choice uses minibatch OT or entropic OT. A VAE-like choice learns $q_\phi(z \mid x)$ and induces $\Gamma_\phi(\mathrm{d}z, \mathrm{d}x) = q_\phi(\mathrm{d}z \mid x)\pi_{\text{data}}(\mathrm{d}x)$. The coupling controls semantic alignment and path complexity.

2. Learn or choose the bridge. A deterministic bridge gives flow matching. A Brownian bridge gives diffusion flow matching. A Schrödinger bridge optimizes the bridge under entropy regularization. A Poisson/electrostatic bridge chooses an augmented field geometry. A learnable bridge would optimize λ jointly with the generator, analogous to learning the encoder family in a VAE.

3. Project the bridge onto a causal Markov decoder. Given samples from the bridge and a local target u_t or β_t, train

$$\min_\theta \int_0^1 \mathbb{E}\left[\|u_t - v_\theta(X_t, t)\|^2\right] \mathrm{d}t \tag{7.1}$$

for an ODE gauge, or

$$\min_\theta \int_0^1 \mathbb{E}\left[\|\sigma_t^{-1}(\beta_t - b_\theta(X_t, t))\|^2\right] \mathrm{d}t \tag{7.2}$$

for a shared-covariance SDE gauge.

4. Choose a gauge for sampling. Once a marginal path and current are learned, choose whether to sample with an ODE, with an SDE, or with an augmented-space field. This choice may trade off likelihood tractability, mode coverage, numerical stability, robustness, and guidance behavior.

8 Guiding principles

The proposed view suggests several practical principles.

Bridge first, sampler second. The bridge family determines the inductive bias. Straight-line rectified flow, Brownian diffusion, entropic Schrödinger transport, and Poisson harmonic fields are different answers to the same question: which latent path posterior should the model imitate?

The score is a gauge correction. The score appears when one represents a probability current with nonzero diffusion. Removing the diffusion gives an ODE representation; adding diffusion requires a score correction. This is why ODE and SDE samplers can share marginals.

Coupling is representation learning. In independent coupling, the model must learn both semantic alignment and local dynamics. In encoder-induced or OT-induced coupling, some alignment is moved into inference. This is analogous to the role of an encoder in a VAE.

Poisson flow is a bridge geometry, not an exception. The electric field is a way to define paths and currents in augmented space. It should be compared to Brownian, OT, and stochastic-interpolant bridges as a geometry choice.

A full theory should optimize bridge, coupling, and gauge jointly. Most current models fix at least two of these choices. A more general model would learn the endpoint coupling, bridge geometry, Markov decoder, and gauge under a single variational objective.

9 Open problems

Learnable bridge posteriors. Can we parameterize $\mathsf{R}_\lambda^{z,x}$ richly enough to improve sample quality and likelihood while keeping bridge sampling and target computation tractable?

Gauge selection. Given a learned current J_t, how should one choose the diffusion matrix a_t for best numerical stability, guidance performance, or likelihood? PFGM++ suggests that intermediate gauges between pure Poisson flow and diffusion can be beneficial (17).

Path-space amortization. A VAE-style model should amortize not only endpoint latent variables but also path variables. Latent Stochastic Interpolants are a concrete step in this direction (21). More general amortized bridges could connect representation learning with flow/diffusion training.

Field/flow equivalence. Recent flow/field matching duality results raise a natural question: which augmented-field geometries are merely reparameterized conditional probability paths, and which are genuinely more expressive (20)?

Beyond Brownian reference processes. Brownian bridges are mathematically convenient, but not necessarily the best bridge for data with memory, hierarchy, or manifold constraints. Fractional, kinetic, Hamiltonian, or learned reference processes may provide better path priors.

10 Conclusion

This preliminary manuscript proposes a path-space variational bridge as a unifying mathematical object for diffusion models, rectified flows, flow matching, Brownian/Schrödinger bridges, and Poisson/electrostatic flows. The framework has four separable ingredients: endpoint coupling, bridge path measure, Markovian projection, and probability-current gauge. Under this lens, diffusion models are path-space VAEs with noisy encoder bridges; rectified flows are deterministic bridge projections; Brownian and Schrödinger bridges are entropic stochastic bridge choices; and Poisson flows are harmonic-field bridge geometries in augmented space. The most promising next step is to make the bridge itself learnable under a continuous-time ELBO while preserving the computational simplicity of matching losses.

Appendix A. Proofs

A.1 Projection loss

For any measurable vector field v, expand Eq. (4.3) pointwise in (x, t) using the conditional law of u_t given $X_t = x$:

$$\mathbb{E}[\|u_t - v(X_t, t)\|^2] = \mathbb{E}\left[\mathbb{E}[\|u_t - v(X_t, t)\|^2 \mid X_t]\right] \tag{A.1}$$

$$= \mathbb{E}\left[\mathrm{Var}(u_t \mid X_t) + \|\mathbb{E}[u_t \mid X_t] - v(X_t, t)\|^2\right]. \tag{A.2}$$

The variance term is independent of v. The unique L^2 minimizer is therefore $v^\star(x, t) = \mathbb{E}[u_t \mid X_t = x]$.

A.2 Probability-current gauge

The Fokker-Planck equation for Eq. (5.2) is

$$\partial_t \rho_t = -\sum_i \partial_i(b_i \rho_t) + \frac{1}{2}\sum_{i,j} \partial_i \partial_j (a_{ij}\rho_t). \tag{A.3}$$

Define

$$J_i = \rho_t b_i - \frac{1}{2}\sum_j \partial_j(a_{ij}\rho_t). \tag{A.4}$$

Then $\partial_t \rho_t + \nabla \cdot J = 0$. If $b = b^{(a)}$ as in Eq. (5.4), then

$$J_i = \rho_t v_i + \frac{1}{2}\sum_j \partial_j(a_{ij}\rho_t) - \frac{1}{2}\sum_j \partial_j(a_{ij}\rho_t) = \rho_t v_i. \tag{A.5}$$

Thus the SDE and ODE share the same probability current and hence the same marginal density path, under uniqueness of the Fokker-Planck solution.

Appendix B. A minimal experimental program

A practical version of this paper could start with four small experiments.

1. Compare independent, OT, and encoder-induced couplings under the same linear bridge and ODE gauge.
2. Compare deterministic, Brownian, and electrostatic bridge geometries under the same neural architecture and compute path curvature, likelihood proxy, and sample quality.
3. Fix a learned probability path and vary the diffusion gauge a_t to measure stability and robustness.
4. Implement a latent path ELBO by learning $q_\phi(z \mid x)$ jointly with a latent stochastic interpolant and decoder, then compare with a fixed autoencoder plus latent flow matching.

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